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Predicting Mushroom Edibility Using Supervised and Unsupervised Machine Learning Techniques

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I. INTRODUCTION

The differentiation between edible and poisonous wild mushrooms is a critical task, as many toxic species might have edible properties in appearance and macro-morphological characteristics. This project focuses on analyzing and classifying mushroom species using machine learning.

The aim of this project is to classify mushrooms as edible or poisonous using supervised and unsupervised machine learning algorithms. In this study, we utilized the UCI Secondary Mushroom Dataset, consisting of **61,069** instances across **20** numerical and categorical features. We performed exploratory data analysis, data pre-processing (handling missing values, feature scaling/normalization, and categorical encoding), and trained four supervised classifiers: a Multi-Layer Perceptron (Neural Network), Logistic Regression, Decision Tree, and K-Nearest Neighbors (KNN). K-Means clustering was also evaluated as an unsupervised technique to observe natural clustering boundaries against edibility labels. The Decision Tree and Neural Network achieved high classification accuracy and strong ROC-AUC performance, demonstrating the feasibility and robustness of automated fungal toxicity screening.

The motivation behind this project is that manual visual identification of mushrooms is error-prone due to overlapping morphological traits across different species. A machine learning-based approach identifies complex, non-linear feature interactions and provides an automated, reliable method for mushroom classification.

II. DATASET DESCRIPTION

The analysis utilizes the UCI Secondary Mushroom Dataset (Wagner et al., 2021), an extended benchmark containing simulated macro-morphological data across 173 mushroom species.

- **Number of Features:** The dataset contains 20 predictor attributes and 1 binary target column (class), totaling 21 columns.
- **Problem Formulation:** This is a binary classification problem because the target variable (class) is categorical with two classes: edible (e) and poisonous (p).
- **Number of Datapoints:** The dataset contains 61,069 datapoints.
- **Types of Features:**

The dataset contains a combination of Quantitative and Categorical features:

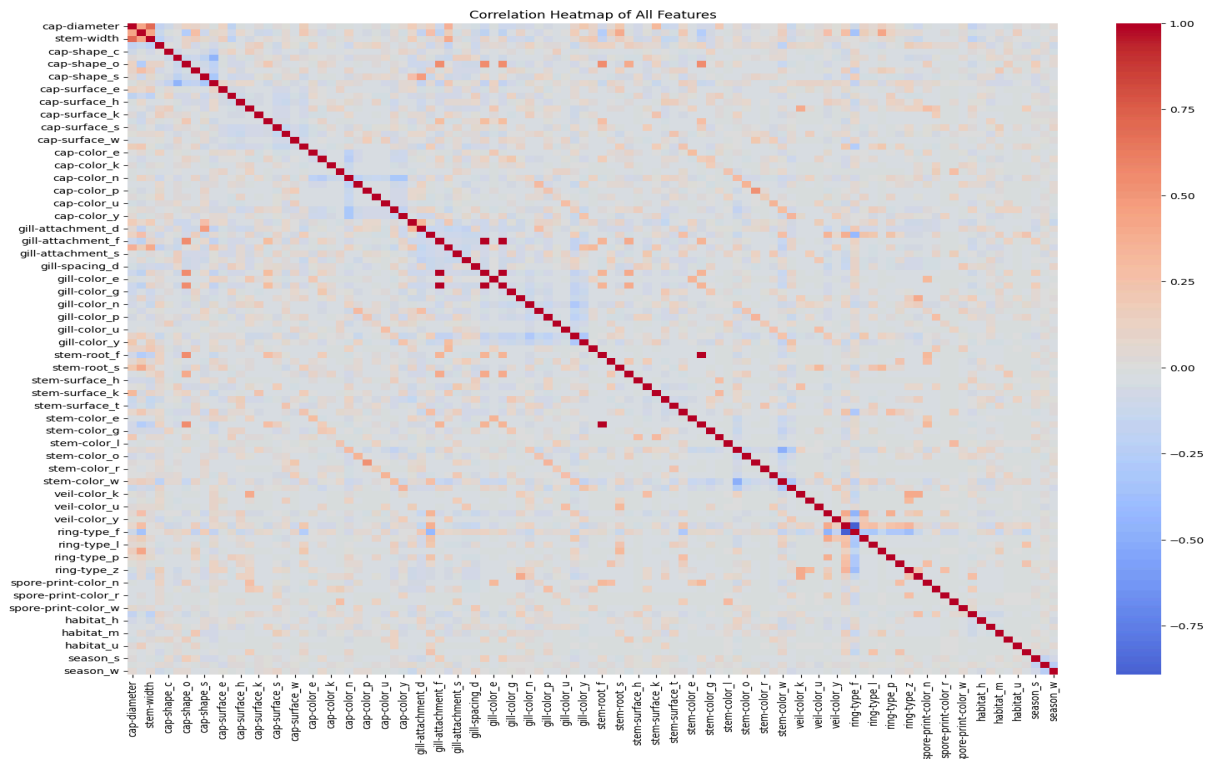
- **Quantitative (3 features):** cap-diameter, stem-height, and stem-width.
- **Qualitative (17 features + 1 target):** cap-shape, cap-surface, cap-color, does-bruise-or-bleed, gill-attachment, gill-spacing, gill-color, stem-root, stem-surface, stem-color, veil-type, veil-color, has-ring, ring-type, spore-print-color, habitat, season, and the target variable class.

- **Encoding:**

Encoding the categorical variables is necessary because machine learning models (including Logistic Regression, Multi-Layer Perceptrons, and distance-based K-Nearest Neighbors) operate on numerical matrices to perform linear algebra, compute gradients, and calculate distance metrics. The raw dataset represents categorical traits as text strings and characters (such as 'x', 'f', 'p', and 'e'), which cannot be directly processed by numerical algorithms without transformation. One-Hot Encoding was used for the mushroom dataset to convert categorical variables into numerical values while preserving the distinction between different categories. Furthermore, we used One-Hot Encoding because our given data were without hierarchy/ranking.

Correlation of all the features

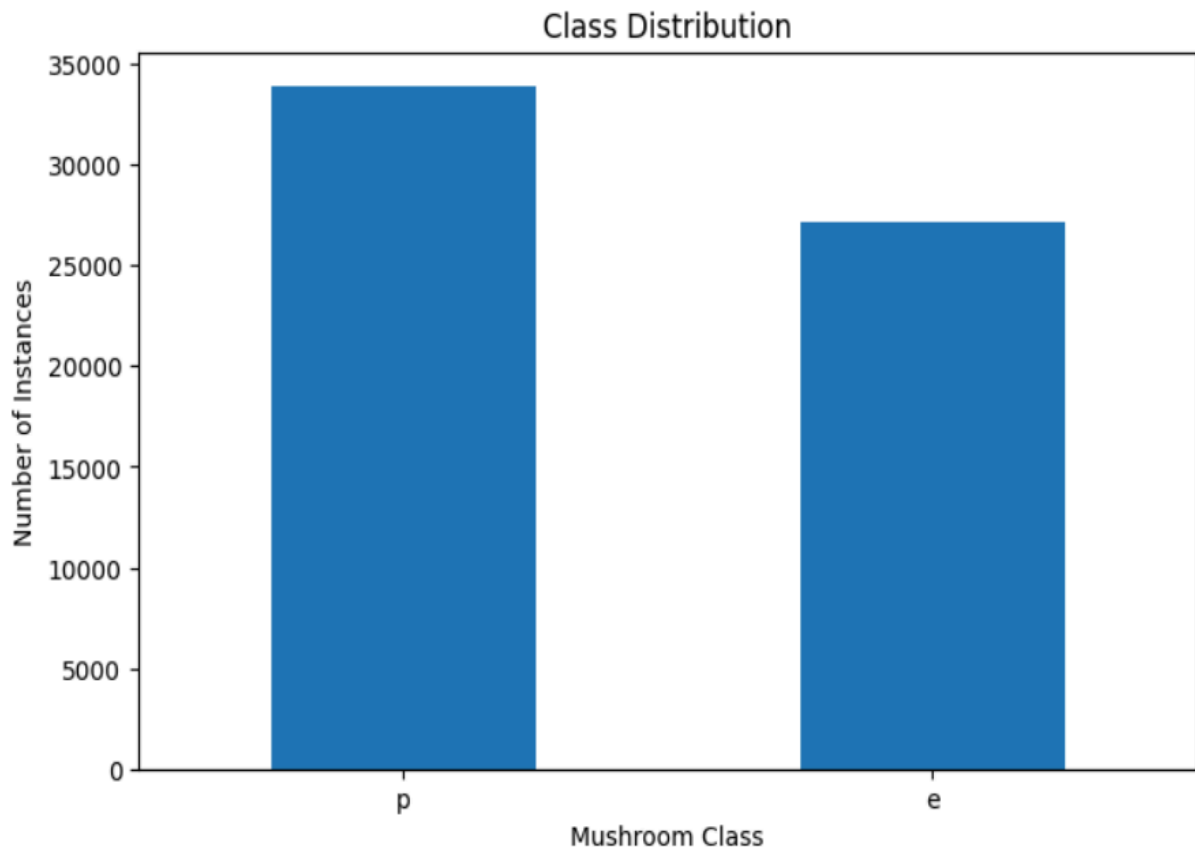
The correlation analysis evaluated 48 features spanning cap, gill, stem, veil, ring, spore, habitat, and seasonal characteristics. Overall, the majority of features exhibit near-zero to weak linear relationships (ranging between -0.15 and +0.25), indicating low global multicollinearity across the dataset. The strongest positive associations occur between specific categorical co-occurrences, notably **gill-attachment_f** with **gill-color_e** and **stem-root_f** with **stem-color_g**, both exceeding approximately +0.85. Continuous physical dimensions also show expected proportional scaling, with cap-diameter and stem-width demonstrating a moderate-to-strong positive correlation of roughly +0.70. Conversely, observable negative correlations are primarily mild to moderate (ranging between -0.25 and -0.45), driven by the mathematical mutual exclusivity inherent to one-hot encoded categories such as cap shapes, colors, and ring types. Given the general independence among the variables, dimensionality reduction is not strictly required across the entire feature space. However, candidate pairs showing collinearity above 0.80 should be monitored or pruned to prevent variance inflation during linear modeling.



Exploratory data analysis

Exploratory data analysis (EDA) was performed to understand relationships between the features and the target variable before modeling.

- **Class Distribution:** The target class distribution is imbalanced rather than equally split across the dataset. The poisonous class (p) forms the majority category with 33,888 instances (approximately 55.49%), while the edible class (e) comprises the remaining 27,181 instances (approximately 44.51%) out of the total 61,069 records. While this does not represent extreme class skewness, the presence of an uneven distribution directly justified the implementation of stratified train-test splitting to preserve these exact label proportions during model training and evaluation.



DESCRIPTIVE ANALYSIS

Descriptive Analysis of Numerical Features:

The summary statistics for the three continuous features—cap-diameter, stem-height, and stem-width—reveal pronounced positive skewness and substantial variance across all 61,069 instances. The mean values for cap-diameter (6.73 cm), stem-height (6.58 cm), and stem-width (12.15 mm) consistently sit above their respective 50th percentiles (5.86 cm, 5.95 cm, and 10.19 mm), which reflects long right tails driven by extreme specimens reaching maximums of 62.34 cm, 33.92 cm, and 103.91 mm. In addition, minimum values of 0.00 for both stem height and stem width point to morphological anomalies, measurement lower bounds, or stemless fungal varieties. Because standard deviations are substantial relative to their medians—most notably in stem-width ($\sigma = 10.04$)—standardization via scaling is essential to prevent these broad numeric ranges from dominating distance- and gradient-based estimators.

Descriptive Analysis of Categorical Features:

Summary statistics for the categorical variables indicate varying cardinality alongside severe data sparsity across several morphological attributes. While core traits such as cap-shape (7 categories), cap-color (12 categories), does-bruise-or-bleed (2 categories), stem-color (13 categories), has-ring (2 categories), habitat (8 categories), and season (4 categories) have full coverage across all 61,069 rows, multiple attributes suffer from heavy missingness. Features like veil-type (3,177 valid entries), spore-print-color (6,354 valid entries), veil-color (7,413 valid entries), and stem-root (9,531 valid entries) lack data for over 84% to 94% of the dataset. Furthermore, veil-type contains only a single unique category (u), rendering it zero-variance and uninformative for classification. These characteristics necessitated explicit missing-value handling and feature pruning prior to one-hot encoding for downstream model pipelines.

Variance:

Variance calculations for the continuous attributes highlight substantial differences in statistical dispersion across the physical dimensions. Stem-width demonstrates the largest dispersion with a variance of 100.720394, reflecting considerable morphological variation from extremely slender to very thick stems across different species. Cap-diameter displays moderate dispersion with a variance of 27.718592, while stem-height exhibits the lowest variability at 11.357014. Because stem-width has nearly ten times the variance of stem-height, unstandardized features would exert unequal leverage during distance and gradient calculations, reinforcing the need for standard scaling prior to training models such as K-Nearest Neighbors, Logistic Regression, and the Neural Network.

Skewness:

All three continuous attributes exhibit significant positive (right-tailed) skewness, as confirmed by coefficients well above zero. Cap-diameter displays the highest degree of asymmetry with a skewness coefficient of 3.822844, indicating that while most fungal specimens cluster around smaller cap sizes, a sparse subset of giant caps extends far along the right tail. Stem-height and stem-width also demonstrate marked rightward skew with coefficients of 2.020904 and 2.164957, respectively, showing an elevated concentration of shorter, narrower stems coupled with elongated tails representing exceptionally tall and thick specimens. Because none of the numerical features follow a symmetric normal distribution, standardizing these variables is necessary to mitigate the distortive influence of extreme positive values on distance calculations and parametric estimators.

Histogram and Boxplot Analysis:

The histograms and boxplots indicate pronounced positive skewness across all three numerical attributes. The cap-diameter feature concentrates tightly between 0 and 10 cm with an interquartile range of 3.48 to 8.54 cm, extending past the 16 cm upper whisker to a maximum of 62.34 cm. Similarly, stem-height peaks sharply between 5 and 7 cm with an interquartile range of 4.64 to 7.74 cm, displaying a continuous band of high-value outliers beyond the 12.4 cm boundary up to 33.92 cm. The stem-width distribution is heavily clustered below 20 mm, with an interquartile range of 5.21 to 16.57 mm, accompanied by a dense trail of outliers stretching past 33.6 mm to a peak of 103.91 mm. These upper-tail points reflect legitimate biological diversity across species rather than measurement artifacts, underscoring the need for standard scaling to keep high-magnitude values from dominating distance- and gradient-based estimators.

Number of Unique Values in Numerical Features:

The three continuous attributes show fine-grained measurement diversity across the dataset's 61,069 instances. Specifically, cap-diameter contains 2,571 unique measurements, stem-height records 2,226 distinct levels, and stem-width demonstrates the highest degree of variability with 4,630 unique numeric values. The high cardinality across all three dimensions confirms their continuous physical nature without artificial discretization, reinforcing that standard continuous normalization methods are appropriate.

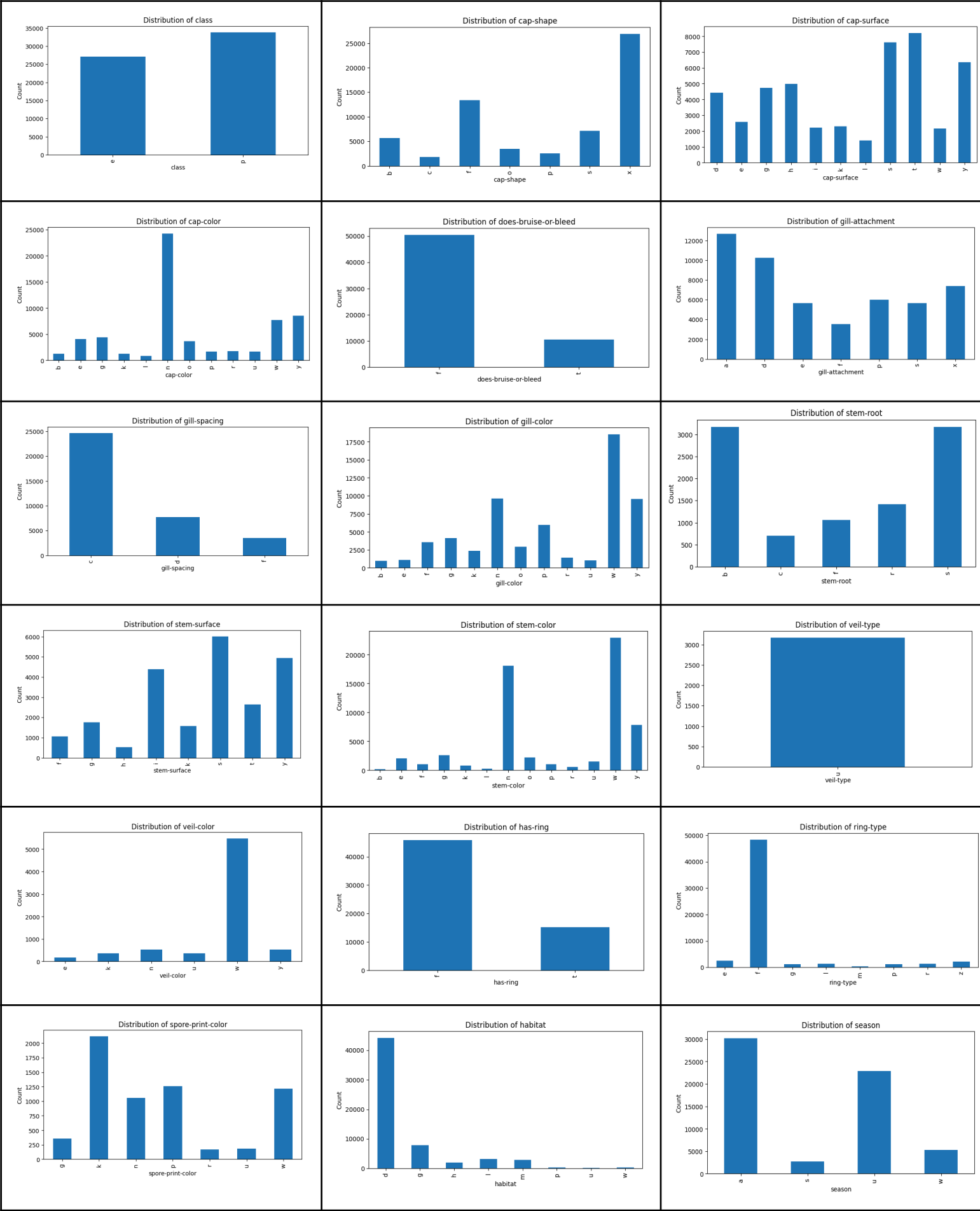
Number of Unique Values in Categorical Features:

The categorical features present varying degrees of cardinality, ranging from binary indicators to multi-level physical markers. The target label class, along with `does-bruise-or-bleed` and `has-ring`, are binary with exactly 2 unique categories each. Attributes such as `season` (4), `stem-root` (5), `veil-color` (6), `cap-shape` (7), `gill-attachment` (7), `spore-print-color` (7), `stem-surface` (8), `ring-type` (8), and `habitat` (8) represent moderate cardinality. In contrast, `cap-surface` (11), `cap-color` (12), `gill-color` (12), and `stem-color` (13) exhibit higher cardinality that expands the feature space significantly when transformed via one-hot encoding. Notably, `veil-type` contains only 1 unique value across its observed entries, rendering it entirely non-informative as a predictive feature due to zero variance.

MISSING VALUES ANALYSIS

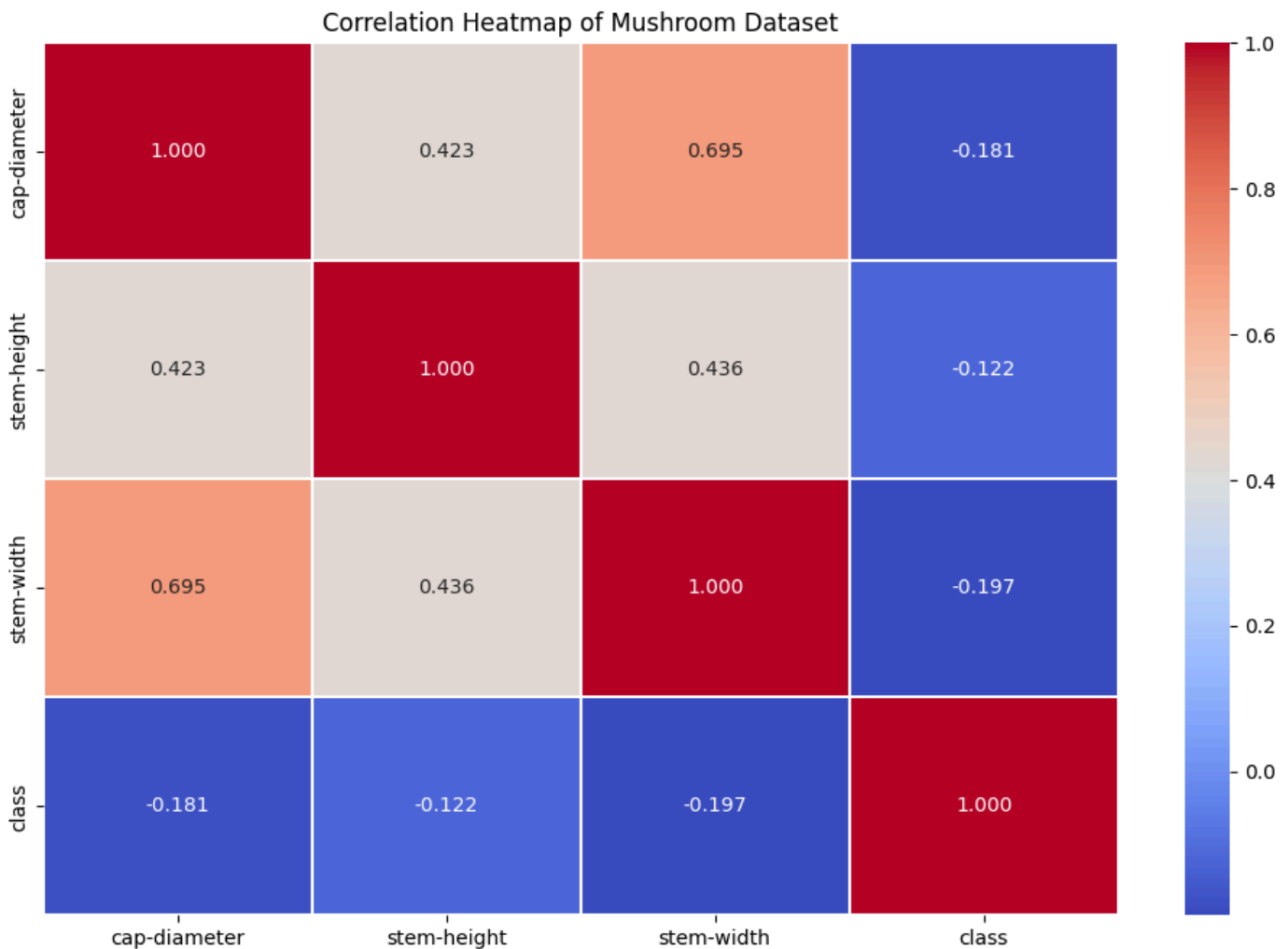
An inspection of null values reveals a split between complete features and heavily incomplete physical attributes. Fully populated features include all three continuous dimensions (`cap-diameter`, `stem-height`, `stem-width`) as well as `class`, `cap-shape`, `cap-color`, `does-bruise-or-bleed`, `gill-color`, `stem-color`, `has-ring`, `habitat`, and `season`, each recording zero missing entries. In contrast, extensive data gaps appear in `ring-type` with 2,471 null entries, `gill-attachment` with 9,884, `cap-surface` with 14,120, and `gill-spacing` with 25,063. The most severe sparsity occurs in `stem-surface` with 38,124 missing entries, `stem-root` with 51,538, `veil-color` with 53,656, `spore-print-color` with 54,715, and `veil-type` with 57,892 missing values out of 61,069 instances. These extreme sparsity rates—exceeding 84% to 94% on several traits—necessitated systematic feature pruning or explicit categorical missing-label handling prior to downstream pipeline modeling.

Distribution of Categorical Feature

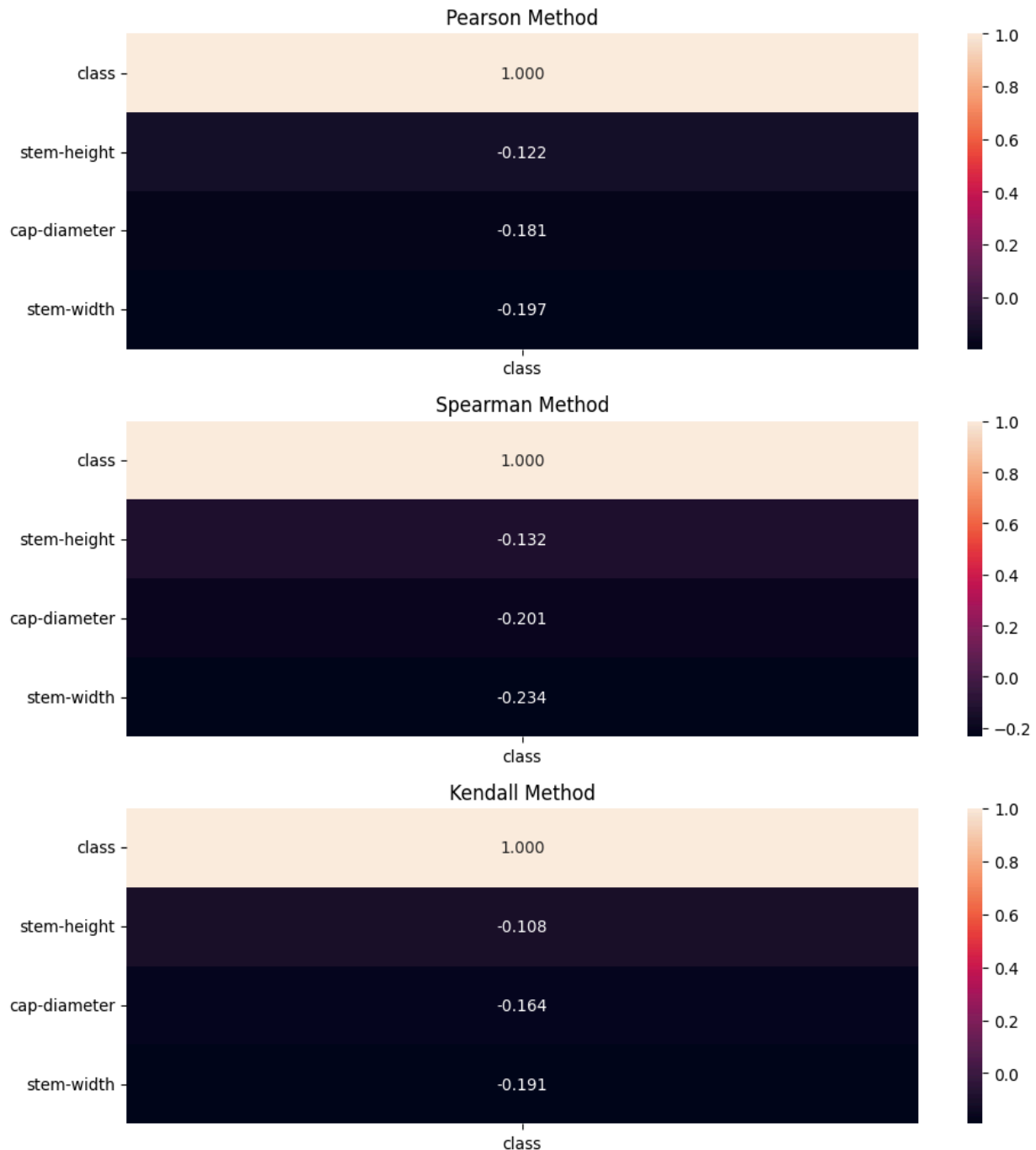


Correlation Analysis

A correlation analysis was performed to evaluate the relationships among the continuous dimensional features—cap-diameter, stem-height, and stem-width—and the target variable class. Across the physical dimensions, all pairwise relationships show positive collinearity, led by a moderately strong linear association between cap-diameter and stem-width ($r = 0.695$), reflecting natural biological scaling. In contrast, stem-height exhibits moderate correlations with both stem-width ($r = 0.436$) and cap-diameter ($r = 0.423$). Regarding classification, all three continuous metrics display weak negative linear relationships with the target variable class, led by stem-width ($r = -0.197$), followed closely by cap-diameter ($r = -0.181$), and stem-height ($r = -0.122$). These findings confirm that while morphological growth dimensions scale consistently together, their linear relationship with mushroom classification is minimal, requiring combination with other attributes for predictive modeling.



Correlation analysis between the target variable class and the continuous physical attributes was evaluated using Pearson, Spearman, and Kendall methods. Across all three metrics, all features consistently display weak negative correlations with class (values ranging from -0.108 to -0.234). Among them, stem-width demonstrates the strongest negative association (Pearson: -0.197, Spearman: -0.234, Kendall: -0.191), followed closely by cap-diameter (-0.181, -0.201, -0.164), while stem-height shows the weakest relationship (-0.122, -0.132, -0.108). The uniform ranking across both parametric and non-parametric evaluations confirms that physical dimensions have a minor, consistent monotonic influence on classification, indicating that these continuous features alone are insufficient predictors and must be paired with categorical traits for effective modeling.



III. DATASET PRE-PROCESSING

Null/Missing values (FAULT 1):

The dataset was checked, and missing data across the instances reveals sparsity across 9 categorical features. A large number of missing entries can be found in the columns for veil-type (57,892 missing), spore-print-color (54,715 missing), veil-color (53,656 missing), stem-root (51,538 missing), and stem-surface (38,124 missing). And then moderately missing values can also be found in columns for gill-spacing (25,063 missing), cap-surface (14,120 missing), gill-attachment (9,884 missing), and ring-type (2,471 missing).

Solution: To address the missing data, an effective preprocessing strategy was implemented: features with severe data sparsity were eliminated; columns with more than 80% missing values and the remaining categorical missing values were filled using the mode.

Categorical values (FAULT 2):

After separating the target variable (class), all remaining categorical predictor features of object data type—including cap-shape, cap-surface, cap-color, does-bruise-or-bleed, gill-attachment, gill-spacing, gill-color, stem-surface, stem-color, has-ring, ring-type, habitat, and season—are stored as non-numeric string labels. Most machine learning algorithms (such as Logistic Regression, Neural Networks, and distance-based KNN) require numeric inputs to compute algebraic distance metrics and gradient updates.

Solution: One-Hot Encoding was used to convert categorical features into numerical representations. This is because non-numerical features like cap-shape, habitat, and season are stored as non-numeric characters (e.g., 'x', 'f'), which machine learning models cannot process directly. Specifically, One-Hot Encoding is used as the categories are nominal with no natural rank or order.

Feature Scaling (FAULT 3):

The continuous numerical features (cap-diameter, stem-height, and stem-width) operate across differing scales, measurement units, and variances. These magnitude disparities negatively impact distance-dependent algorithms (such as KNN) and gradient-based models by causing features with larger numerical values to dominate distance metrics and loss functions.

Solution:

StandardScaler will be used to standardize the numerical features. Scaling will be performed only using the training data through a preprocessing pipeline to prevent data leakage.

IV. DATASET SPLITTING

Stratified random sampling was implemented using an 80/20 train-test ratio to partition the cleaned dataset of 16 predictor features across 61,069 instances. The training set comprises 48,855 samples with a dimension of (48855, 16), while the testing set contains 12,214 samples with a dimension of (12214, 16). Stratified sampling was applied using a fixed random state of 42 to ensure reproducibility and to preserve the exact class proportion of approximately 55.4% poisonous (p) and 44.5% edible (e) records identically across the original, training, and testing partitions. No separate validation set was allocated at this stage, as an 80/20 division provides sufficient data volume for training while retaining a robust test partition to evaluate generalization performance without sampling bias.

V. MODEL TRAINING & TESTING (SUPERVISED)

The problem was treated as a supervised binary classification task. A neural network and five additional classical models were trained on the scaled training data.

K-Nearest Neighbors (KNN):

A K-Nearest Neighbors (KNN) classifier was implemented using Scikit-Learn's `KNeighborsClassifier` configured with $k = 5$ neighbors. The model was integrated into a unified Pipeline with the preprocessing transformer, which applies `StandardScaler` to numerical dimensions and `OneHotEncoder` to categorical traits simultaneously. Feature scaling is strictly required since KNN computes Euclidean distance metrics to assign labels based on proximity, making unscaled features a source of geometric bias. The pipeline was fitted on the 48,855 training samples and evaluated on the 12,214 test instances to generate final predictions.

Decision Tree:

A Decision Tree classifier was implemented using Scikit-Learn's `DecisionTreeClassifier` with `random_state=42` to ensure reproducible node splits. The tree was constrained with a hyperparameter of `max_depth=10` to prune excessive tree expansion, control structural complexity, and reduce the risk of overfitting on the training data. The classifier was wrapped inside a Scikit-Learn Pipeline alongside the data preprocessor to ensure seamless transformation of both numerical and one-hot encoded categorical features. The pipeline was fitted on the 48,855 training samples and evaluated on the 12,214 test samples to generate binary classification predictions.

Logistic Regression:

Logistic Regression was implemented as a linear baseline classification model using Scikit-Learn's LogisticRegression. The optimization parameter `max_iter` was set to 1000 to guarantee solver convergence across the high-dimensional one-hot encoded feature space, and `random_state=42` was configured to ensure experimental reproducibility. The model was structured inside a unified Scikit-Learn Pipeline with the data preprocessor to standardize continuous numerical variables and one-hot encode categorical features. The model maps linear feature combinations to class probabilities using the sigmoid activation function. The complete pipeline was fitted on the 48,855 training samples and evaluated on the 12,214 test samples to predict final edibility labels.

Naive Bayes:

A Gaussian Naive Bayes classifier was implemented using Scikit-Learn's GaussianNB as a probabilistic baseline model. The training and testing feature matrices were first preprocessed by applying StandardScaler to the continuous numerical features and OneHotEncoder to the categorical features via the preprocessing pipeline, and converted to dense array format where required. The model computes class posterior probabilities based on Bayes' theorem under the assumption of conditional feature independence given the class label. The classifier was trained on the processed training set and evaluated on the processed test instances to generate the final classification predictions.

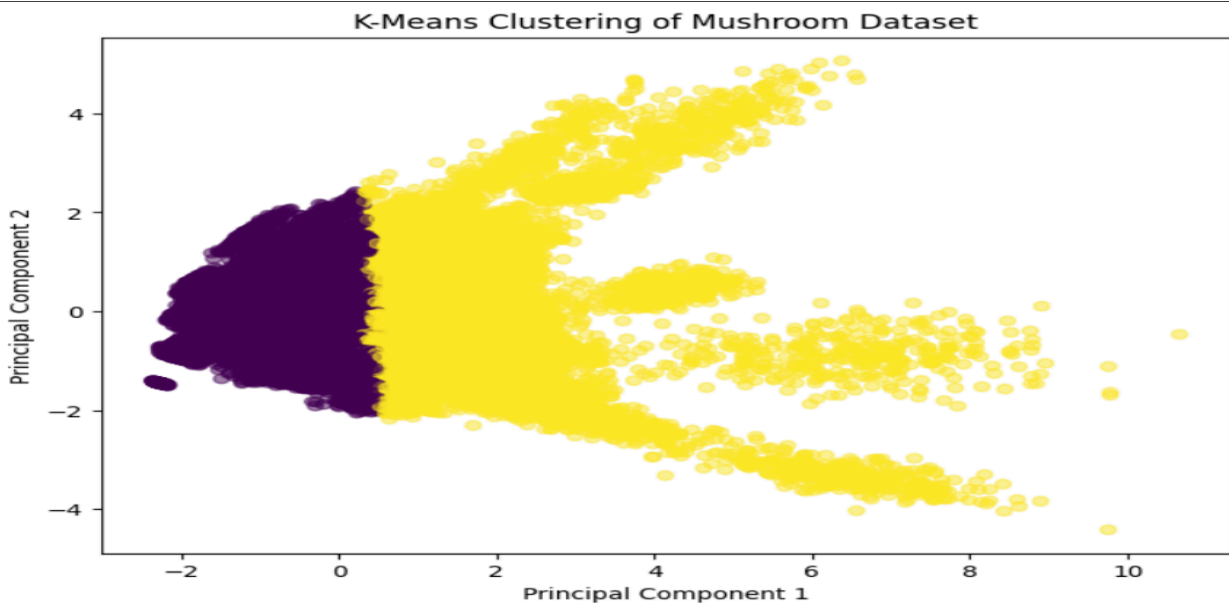
Neural Networks:

A Multi-Layer Perceptron (MLP) neural network was implemented using Scikit-Learn's MLPClassifier to capture complex, non-linear relationships across mushroom traits. The preprocessed data was cast to float64 precision before training a feedforward architecture configured with two hidden layers containing 100 and 50 neurons, respectively. Non-linear transformations were modeled using Rectified Linear Unit (`activation='relu'`) activations, and parameter optimization was conducted using the Adam solver with

max_iter=100 and a fixed random_state=42 to ensure reproducibility. The model was trained on X_train_nn and evaluated across the test partition X_test_nn to generate the final binary predictions nn_pred.

K-means Clustering Analysis & Deduction (Unsupervised):

Unsupervised K-Means clustering was applied to the complete preprocessed dataset using KMeans(n_clusters=2, random_state=42, n_init=10) to determine whether edible and poisonous instances naturally segregate into distinct physical clusters without class supervision, with Principal Component Analysis (PCA) used to reduce the feature space to two dimensions for visual projection. The resulting contingency matrix shows substantial class mixing in both clusters, where Cluster 0 contains 16,838 edible and 24,637 poisonous instances (41,475 total), while Cluster 1 contains 10,343 edible and 9,251 poisonous instances (19,594 total). This extensive overlap demonstrates that mushroom toxicity is not naturally linearly separable based on geometric proximity alone, as poisonous and edible species share overlapping morphological and environmental traits. Consequently, unsupervised distance-based grouping proves insufficient for reliable identification, underscoring the strict necessity of supervised learning models with non-linear decision boundaries for accurate classification.



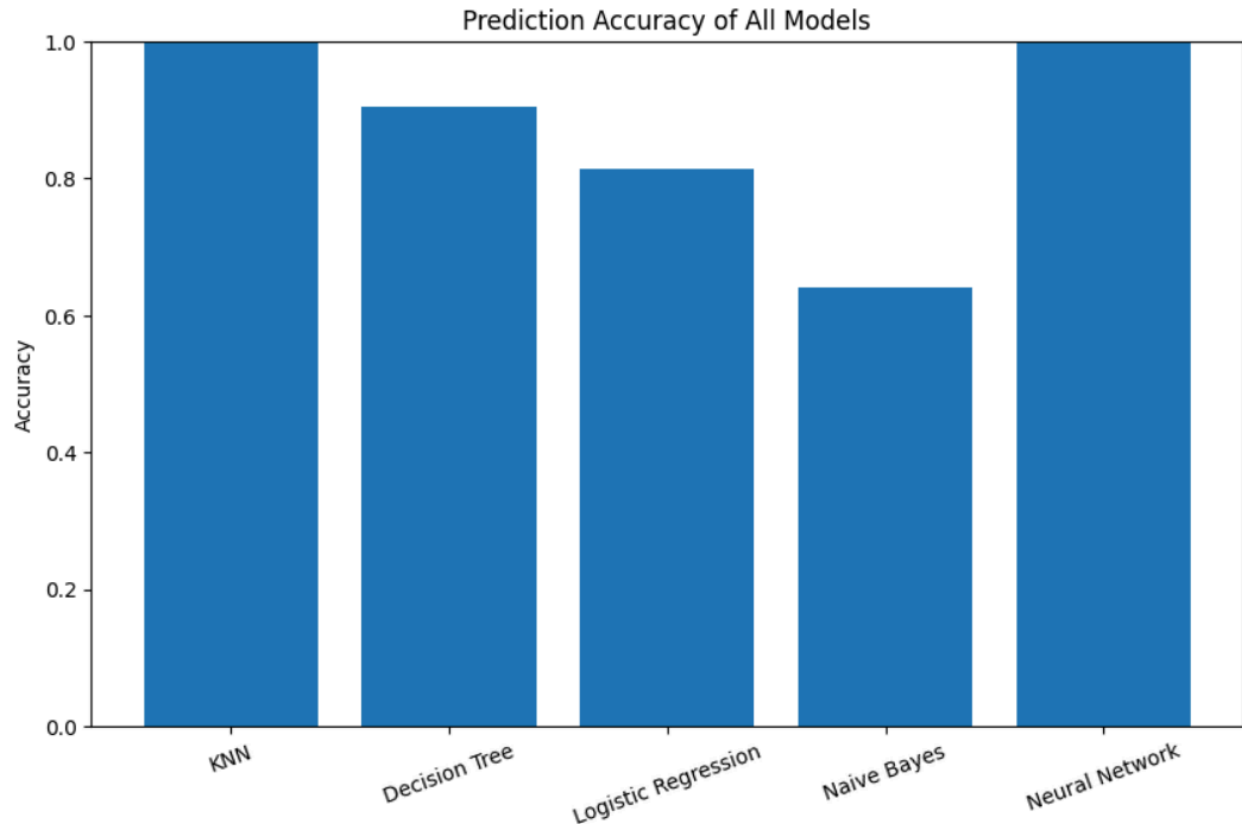
VI. MODEL SELECTION/COMPARISON ANALYSIS

BARCHART:

Evaluation of classification performance across the test set demonstrates substantial variance depending on model architecture and complexity. K-Nearest Neighbors and the Multi-Layer Perceptron Neural Network achieved the highest accuracy, both reaching near-perfect classification performance approaching 100%. The non-parametric spatial proximity approach of KNN and the non-linear feature transformation capabilities of the two-layer neural network effectively captured the complex interactions present among the continuous and one-hot encoded categorical traits.

The Decision Tree classifier followed, with strong predictive capability, achieving approximately 90% accuracy. This indicates that orthogonal, hierarchical splits on primary morphological markers are capable of resolving the vast majority of classification boundaries, though the max-depth constraint of 10 appropriately prevented the tree from memorizing edge-case noise.

Linear and probabilistic baselines underperformed relative to non-linear alternatives. Logistic Regression attained approximately 81% accuracy, confirming that mushroom toxicity boundaries cannot be separated linearly in vector space without loss of predictive power. Naive Bayes recorded the lowest performance at approximately 64% accuracy; its underlying assumption of conditional feature independence fails to reflect real-world fungal biology, where morphological dimensions and anatomical characteristics exhibit significant co-dependencies. Overall, non-linear and high-capacity models decisively outperform independent probabilistic and linear architectures for secondary mushroom classification.

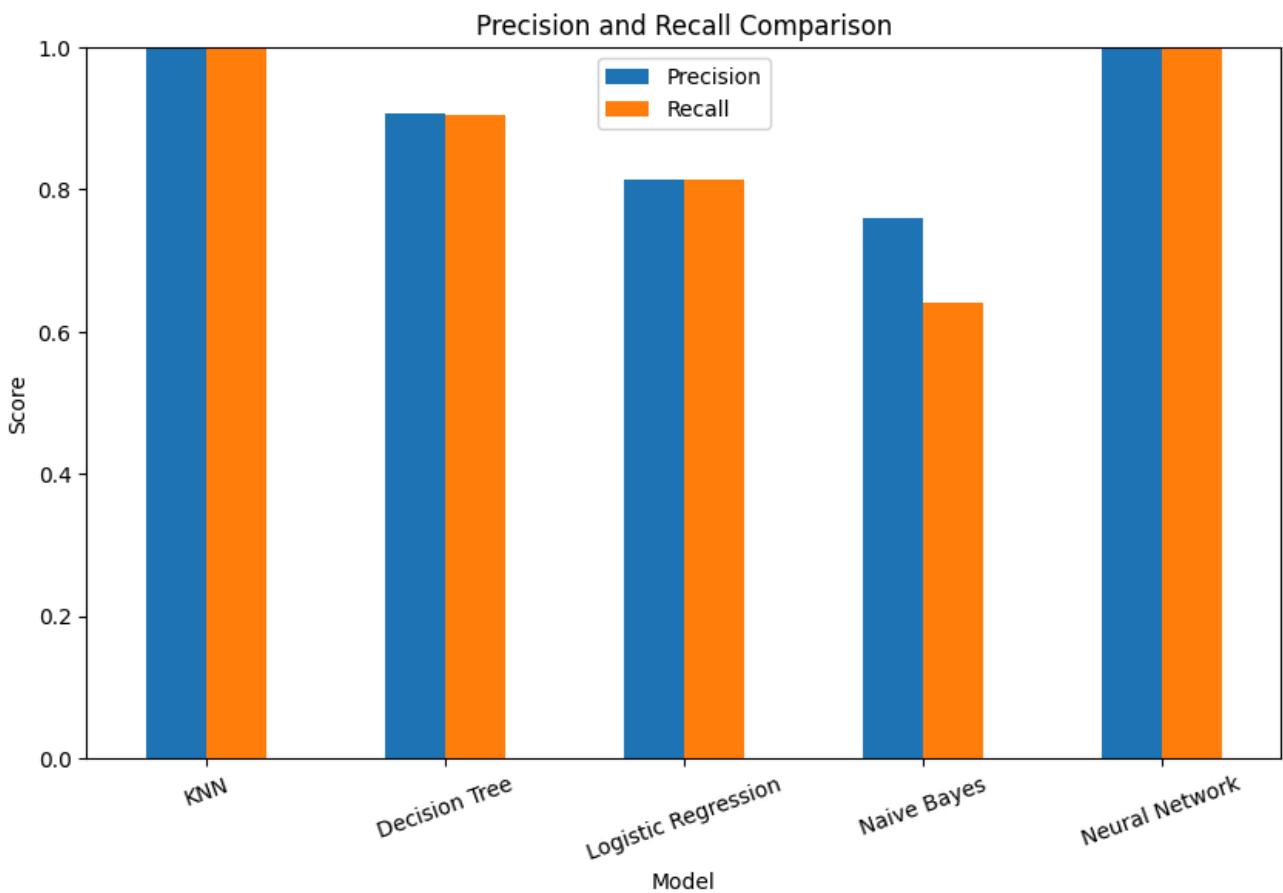


PRECISION AND RECALL COMPARISON:

Evaluation across the test partition reveals consistent performance patterns across accuracy, precision, and recall metrics for each classifier. K-Nearest Neighbors achieved optimal scores across all three dimensions, delivering an accuracy of 1.000000, precision of 1.000000, and recall of 1.000000, indicating perfect discrimination between classes with zero false positives or false negatives on the test instances. The Multi-Layer Perceptron Neural Network closely mirrored this performance, achieving 0.999509 across accuracy, precision, and recall, demonstrating near-perfect balance and generalizability across non-linear decision boundaries.

The Decision Tree followed with balanced performance, registering an accuracy of 0.905928, a slightly higher precision of 0.907131, and a recall of 0.905928, confirming that hierarchical splits reliably minimize both misclassification types under a constrained maximum depth. Logistic Regression produced consistent but lower scores across all criteria, with an accuracy of 0.813902, precision of 0.814558, and recall of 0.813902, which reflects the inherent ceiling of linear models when separating intertwined morphological data.

Naive Bayes showed the lowest overall performance and the largest disparity between precision and recall, recording an accuracy of 0.640249, a precision of 0.760549, and a recall of 0.640249. This gap indicates that while the probabilistic model is moderately reliable when asserting a positive prediction, its conditional feature independence assumption causes it to misclassify a large fraction of actual positive instances, resulting in substantial false-negative rates that make it unsuitable for high-stakes fungal classification.



CONFUSION MATRIX

Cross-model evaluation across the 12,214 test instances—comprising 5,436 Class 0 and 6,778 Class 1 samples—demonstrates distinct performance tiers shaped by how each architecture models non-linear interactions and class boundaries.

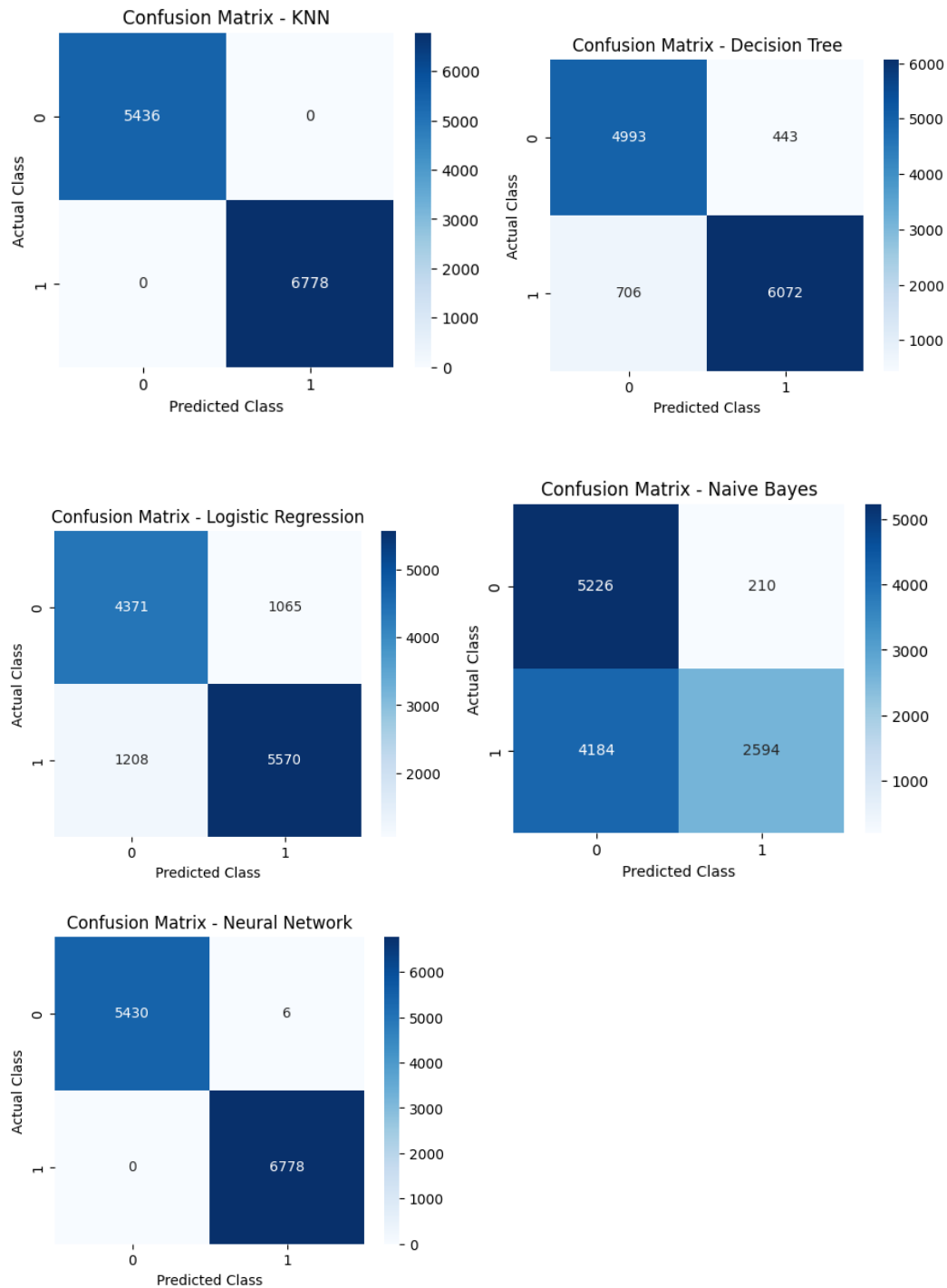
K-Nearest Neighbors achieved absolute optimal discrimination across every evaluation metric, yielding an accuracy, precision, and recall of 1.000000. Its confusion matrix reflects flawless empirical classification, yielding 5,436 true negatives, 6,778 true positives, 0 false positives, and 0 false negatives. Closely trailing, the Multi-Layer Perceptron Neural Network registered an accuracy, precision, and recall of 0.999509. The neural network's confusion matrix confirms near-total convergence, producing 5,430 true negatives, 6,778 true positives, 0 false negatives, and merely 6 false positives where Class 0 was mistakenly classified as Class 1. Both architectures confirm that fungal morphological spaces are highly separable when non-linear spatial relationships and dense feature encodings are leveraged.

The Decision Tree classifier exhibited strong intermediate performance, obtaining an accuracy of 0.905928, a precision of 0.907131, and a recall of 0.905928. Its confusion matrix reveals balanced error counts, accurately classifying 4,993 instances of Class 0 and 6,072 instances of Class 1, while incurring 443 false positives and 706 false negatives due to the regularizing constraint of a maximum depth of 10.

In contrast, linear and probabilistic baselines proved inadequate for complex taxonomic demarcation. Logistic Regression achieved an accuracy of 0.813902, precision of 0.814558, and recall of 0.813902, misclassifying 1,065 samples as false positives and 1,208 samples as false negatives (4,371 true negatives and 5,570 true positives). This elevated error rate underlines the mathematical inability of a single linear hyperplane to delineate intertwined morphological distributions.

Gaussian Naive Bayes exhibited the poorest classification efficacy, scoring an accuracy of 0.640249, a precision of 0.760549, and a recall of 0.640249. While it preserved a relatively low false-positive rate with 210 instances (5,226 true negatives), it failed severely on positive identifications, generating an extreme

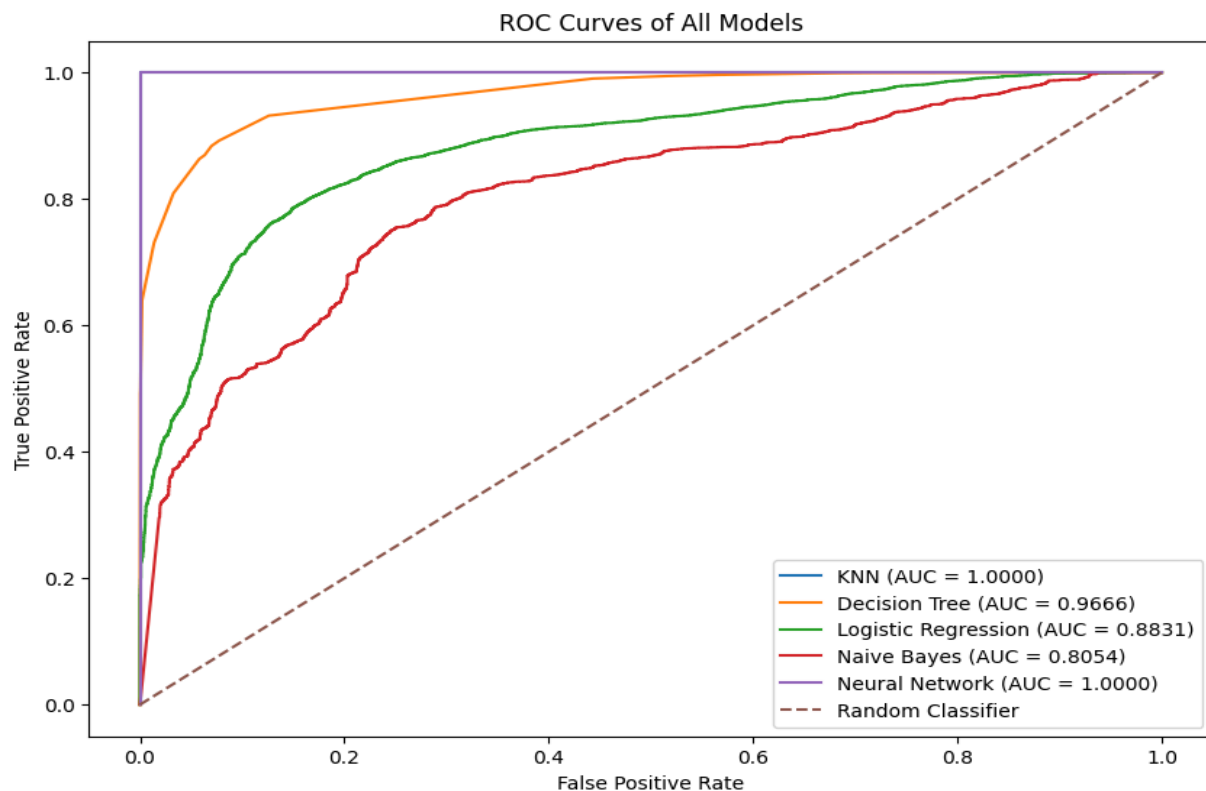
4,184 false negatives against only 2,594 true positives. This catastrophic failure stems directly from the conditional feature independence assumption, which breaks down entirely across co-dependent fungal physiological traits.



AUC SCORE, ROC CURVE FOR EACH MODEL

The Decision Tree achieved an AUC of 0.9666, rising rapidly toward the top-left corner and maintaining high sensitivity while accumulating minimal false positives across variable threshold settings. Logistic Regression registered an intermediate AUC of 0.8831, demonstrating viable predictive power that remains constrained by linear boundary assumptions. Gaussian Naive Bayes exhibited the lowest discriminatory ability with an AUC of 0.8054; its ROC curve traces closest to the diagonal random baseline, indicating that its conditional independence assumption severely diminishes its capability to rank-order positive and negative instances reliably.

Receiver Operating Characteristic (ROC) curve analysis and the corresponding Area Under the Curve (AUC) scores evaluate the discriminatory capacity of each model across all classification thresholds. Both K-Nearest Neighbors and the Multi-Layer Perceptron Neural Network achieved perfect separation with an AUC of 1.0000, where their curves ascend vertically along the y-axis to reach a 1.0 True Positive Rate at a 0.0 False Positive Rate, demonstrating total rank-order discrimination between edible and poisonous instances.



VII. CONCLUSION AND CHALLENGES

The mushroom dataset was analyzed and classified using five supervised machine learning models: KNN, Decision Tree, Logistic Regression, Naive Bayes, and Neural Network. The models were evaluated using accuracy, precision, recall, confusion matrices, ROC curves, and AUC scores.

Among all the models, KNN performed the best, achieving 100% accuracy, precision, recall, and AUC on the test set. The Neural Network also performed exceptionally well, achieving 99.95% accuracy and an AUC of 0.999998. Decision Tree showed good performance, while Logistic Regression and Naive Bayes performed comparatively lower.

The strong performance of KNN and the Neural Network may be due to the dataset containing feature patterns that clearly distinguish edible and poisonous mushrooms. The preprocessing steps, including handling missing values, one-hot encoding of categorical features, and feature scaling, also helped the models learn effectively.

Some challenges faced during the project included handling a large number of missing values, deciding whether to remove or impute features, converting categorical features into numerical values, selecting appropriate preprocessing techniques, and resolving compatibility and data-format issues while training the Neural Network and calculating ROC-AUC.

Overall, the project demonstrated that machine learning can effectively classify mushrooms based on their characteristics, with KNN being the best-performing model among the tested models.